

Extra Neural Metastasis Of Glioblastoma Multiforme: A Literature Review.

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Extra-neural metastases of glioblastoma multiforme are uncommon with unidentified metastatic mechanism. There is no consensus over optimum treatment regimen. The current narrative review was planned to illuminate the presence criteria, sites of metastatic spread, incidence, mechanism, risk factors and management.

Unusual extraneural metastasis of glioblastoma.

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Glioblastoma (GB) is the most common and aggressive malignant brain tumor in adults. Extracranial metastases are very rare, been described in the lungs, soft tissue, or the intraspinal space. Through a PubMed-based bibliographic search, the authors reviewed the cases reported in the literature to date, emphasizing the epidemiology and pathophysiology of this rare condition. A clinical case of a 46-year-old man with an initial diagnosis of gliosarcoma, who received complete surgical and adjuvant treatment and later recurred as GB with incidental finding of a lung tumor, whose pathology reported metastasis of the primary, is illustrated. Understanding the pathophysiology, it is likely that the incidence of extraneural metastases may continue to increase. Considering improvements in diagnostic techniques that allow early diagnosis, as well as advances in neurosurgical therapy and multimodal management with the aim of improving patient survival, the period in which malignant cells can spread and form extracranial metastases could increase. When screening should be performed to detect metastases in these patients is still not clear. The neuro-oncologists should pay attention to the systematic survey for extraneural metastasis of the GB. Timely detection and early treatment improve overall quality of patients' life.

Extracranial Glioblastoma Metastasis: A Neuropathological Case Report.

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Glioblastoma (GBM) is a central nervous system (CNS) high-grade glioma with a dismal patient prognosis. Classical concepts surrounding GBM development and progression indicate its ability to produce metastasis within the CNS, one of the few primary tumors with such capabilities. While classical concepts state that no primary CNS tumor produces extracranial metastasis, there have been multiple reports of such occurrences over the previous two decades. Here, we report a case of a male in his forties who presented to our institution with complaints of progressive headache and a history of right temporal craniotomy one month prior with a histologically verified GBM performed at another institution. Neuroradiology confirmed a residual tumor in the areas of the previous craniotomy, and gross total excision confirmed the diagnosis of GBM, although based on the presence of connective tissue amidst the tumor stroma, gliosarcoma could not be ruled out. The patient initiated treatment, and his condition remained stable for four calendar years until he again presented to our institution with a rapidly growing tumor mass in the right lateral neck region. Excision of the neck mass showed histopathological features of a tumor comprised of atypical cells with pronounced polymorphism, some with spindle cell morphology and a tendency for fascicular growth and focal palisade necrosis. Immunohistochemistry with a broad set of markers disproved epithelial, mesenchymal, melanocytic, and lymphoid genesis, with some markers of

glial genesis present; hence, metastatic GBM was established. The patient reinitiated treatment and is currently stable. The steadily increasing amount of similar reported cases, together with the steady, albeit small, increase in GBM patient survival and improvement of neurooncological healthcare distribution and follow-up, challenge the classical concepts of GBM and other primary CNS tumors being unable to produce metastasis and swaying this perception towards the biological capabilities of these tumors to produce metastasis, while such rarely develop due to the short patient survival.

An Unusual Case of Glioblastoma Multiforme, Presenting as Skeletal Superscan.

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Extracranial metastases of glioblastoma multiforme (GBM) are very rare. The estimated incidence is <2%. We report a case of a 49-year-old woman, who was a known case of GBM in the left temporo-occipital lobe. She was operated and had received radiotherapy and adjuvant chemotherapy for the same. Subsequently, the patient underwent bone scan. On 99 m-Tc methylene diphosphonate (MDP) bone scan, homogenously increased tracer uptake was noted in the axial and appendicular skeletal system, suggesting metastatic skeletal superscan.

Emerging insights into barriers to effective brain tumor therapeutics.

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There is great promise that ongoing advances in the delivery of therapeutics to the central nervous system (CNS) combined with rapidly expanding knowledge of brain tumor patho-biology will provide new, more effective therapies. Brain tumors that form from brain cells, as opposed to those that come from other parts of the body, rarely metastasize outside of the CNS. Instead, the tumor cells invade deep into the brain itself, causing disruption in brain circuits, blood vessel and blood flow changes, and tissue swelling. Patients with the most common and deadly form, glioblastoma (GBM) rarely live more than 2 years even with the most aggressive treatments and often with devastating neurological consequences. Current treatments include maximal safe surgical removal or biopsy followed by radiation and chemotherapy to address the residual tumor mass and invading tumor cells. However, delivering effective and sustained treatments to these invading cells without damaging healthy brain tissue is a major challenge and focus of the emerging fields of nanomedicine and viral and cell-based therapies. New treatment strategies, particularly those directed against the invasive component of this devastating CNS disease, are sorely needed. In this review, we (1) discuss the history and evolution of treatments for GBM, (2) define and explore three critical barriers to improving therapeutic delivery to invasive brain tumors, specifically, the neuro-vascular unit as it relates to the blood brain barrier, the extra-cellular space in regard to the brain penetration barrier, and the tumor genetic heterogeneity and instability in association with the treatment efficacy barrier, and (3) identify promising new therapeutic delivery approaches that have the potential to address these barriers and create sustained, meaningful efficacy against GBM.

A Rare Case of Glioblastoma With Extensive Liver Metastases.

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Glioblastoma (GBM), the most common primary brain tumor, is a highly aggressive malignancy for which the median survival is about 13 months, and 5-year survival is well under 5%. These tumors usually occur in the brain and enlarge and infiltrate through white matter, including crossing through the corpus callosum to the opposite cerebral hemisphere. They may spread to distant parts of the central nervous system (CNS) via cerebrospinal fluid pathways. Extraneural metastases from primary brain tumors are quite rare, for 2 probable reasons: because most patients survive less than 2 years, and because of the absence of true lymphatics in the CNS. Typical sites for distant extraneural metastasis of GBM are lungs and pleura, followed by lymph nodes and bones; spread to the liver is exceptional. Most of the reported cases with liver metastases had either single or only a few such metastatic lesions. We report a probably unique case of GBM with extensive liver metastases along with a review of previous cases of liver metastasis from GBM, and we discuss the possible mechanisms of metastasis.

Extraneural metastases in glioblastoma patients: two cases with YKL-40-positive glioblastomas and a meta-analysis of the literature.

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Glioblastoma (GBM) are high-grade gliomas that severely impact on overall survival (OS). GBM cell motility and the breakdown of the blood-brain barrier could favor GBM cell communication with the systemic circulation. In spite of this, extracranial GBM metastases are rare. Here, we describe two YKL-40-positive GBM patients with extra-CNS (central nervous system) metastases, and we present a meta-analysis of 94 cases. The analysis concluded that extra-CNS metastases occurred 8.5 months after first GBM diagnosis and OS was 12 months; surgical GBM excision was associated at a longer interval to extra-CNS metastasis than biopsy only, and even longer if followed by radiotherapy and chemotherapy. Both our case reports were adult males who developed extra-CNS, YKL-40-positive metastases at lymph nodes, lung and subcutaneous sites, after 86 and 24 months from initial diagnosis of GBM. At first GBM local recurrence, they were treated with bevacizumab (BV), an anti-vascular endothelial growth factor antibody. They died after 4 and 1 month from the occurrence of metastases. Both cases expressed YKL-40 and lacked EGFR amplification, suggesting a mesenchymal phenotype, and maintained such profile at extra-CNS recurrence; they did not show MGMT promoter methylation, IDH1/2 mutations, or c-Met upregulation. Our two cases and the meta-analysis support the idea that prolonged survival of GBM patients increases the probability of GBM cells shedding to lymphatic and hematic system. Interestingly, the present two cases showed the features of mesenchymal profile, usually related with worst prognosis that was maintained in extracranial metastases.

Multiple extraneural metastasis of glioblastoma multiforme.

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Glioblastoma multiforme is the most frequent primary tumor in the brain. Despite improvements in its surgical, chemotherapy and radiotherapy treatment, prognosis remains poor. Extracranial metastases of glioblastoma are a rare complication in this disease. Its appearance has been described in lung, liver, bone or lymph nodes. We describe the case of a 20 year-old patient who complained of a subacute-onset headache. In the MRI an enhancing right temporal lesion was detected suggesting a high grade glioma as

first diagnosis. Surgery was performed, obtaining a gross total resection of the lesion. Our patient underwent adjuvant radiotherapy and chemotherapy treatment, according to our hospital's protocol. Five months after initial surgery our patient complained of chest pain and a hacking cough. A thoracic-abdominal-pelvic CT scan was obtained, which showed bilateral lung infiltrates with pleural effusion, a pancreatic nodule and several vertebral lytic lesions. The lung lesions were biopsied. The pathologic diagnosis was metastatic glioblastoma multiforme. The patient died eight months after initial diagnosis. Extracranial metastases of glioblastoma remain a rare event although its incidence is increasing, probably due to the improvement in survival among these patients and better imaging techniques. The mechanisms for extracranial dissemination of glioblastoma are not entirely known, as several theories exist in this regard. Physicians must be aware of this complication and keep it in mind as a differential diagnosis to improve the quality of life of our patients.

Metastatic glioblastoma: case presentations and a review of the literature.

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Extracranial metastases from glioblastoma (GBM) are uncommon with an estimated incidence of less than 2%. We report two cases of metastatic GBM seen within an 8-week period followed by a literature review. We attempted to identify common factors or a causative mechanism. Factors that predominated among the reviewed cases included male gender, tumor location, and younger age. Causative mechanisms were not apparent. While metastatic disease remains rare, it might be occurring with increasing frequency. This trend might be due to increased diagnosis, better imaging, a more extensive physician workup, or an increase in survival. Metastatic GBM can present and progress quite rapidly, and repeat evaluations of persistent or worsening complaints among GBM patients are warranted. Early diagnosis of metastatic disease spread can help to expedite alleviation of patients' discomfort, in an already aggressive disease process.

The natural history of extracranial metastasis from glioblastoma multiforme.

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Extracranial metastasis is a unique but rare manifestation of glioblastoma multiforme. It is thought to arise from glioblastoma cells disseminated into the blood stream. We undertook a comprehensive analysis of 88 cases of extracranial glioblastoma (5 were gliosarcomas) published between 1928 and 2009. Cases included in the analysis were primary or secondary glioblastomas that subsequently invaded organs outside the brain or spinal cord. The median age was 38 years and the median overall survival time was 10.5 months (range 0.0-60.0 months). The median time from symptom onset to diagnosis of primary glioblastoma was 2.5 months, from diagnosis to detection of extracranial metastasis was 8.5 months, and from metastasis to death was 1.5 months. From 1940 to 2009, there has been progressive lengthening of the interval from detection of extracranial metastasis to death, at a rate of 0.7 months per decade (95% confidence interval 0.5-1.0 month). Use of magnetic resonance imaging correlates with an increase in overall survival but not age, gender, or site of primary glioblastoma. Patients treated with surgery + radiation + chemotherapy + cerebrospinal fluid shunting had the longest average survival interval from metastasis to death when compared to those treated with surgery alone, radiation alone, surgery + radiation, and surgery + radiation + chemotherapy. Lung metastasis is a prognostic factor of extremely poor outcomes. We conclude that patients with glioblastoma extracranial metastasis have

poor prognosis, but there has been a progressive lengthening of survival in each successive decade from 1940 to 2000.
